16th AIR FORCE
Commercial Solutions Opening (CSO) Solicitation
Number: FA7037-23-S-C0001

Date: 8 November 2022

The Government is not obligated to make any award under this CSO and any potential award consideration is subject to availability of funds.

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I. OVERVIEW

The 16th Air Force is offering a CSO authorized by Department of Defense (DoD) Class Deviation (CD) 2022-O0007. Under a CSO, the USAF may competitively award contracts to proposals received in response to a general solicitation, similar to a Broad Agency Announcement (BAA), to acquire innovative commercial items, technologies, and services, based on a review of proposals by scientific, technological, or other subject-matter expert peers within the USAF. Under this CSO, all items, technologies, and services shall be treated as commercial items.

The USAF intends to obtain "innovative" solutions or potential new capabilities that fulfill requirements, close capability gaps, or provide potential technology advancements. Solutions may include existing technologies or procedures, not currently in use by the USAF, that would enhance or streamline USAF mission capabilities. "Innovative" is defined as any technology, process or method that is new as of the date of submission of a proposal. It also includes any new application of an existing technology, process, or method.

This is a CSO with Calls. This solicitation *is not* requesting any white papers or proposals at this time. Further, unsolicited proposals will not be accepted. Funding is not available at this time. It is the responsibility of the potential responders to monitor this site for additional information pertaining to this CSO. This CSO is the overarching umbrella solicitation from which Calls will be issued by AMIC Det 2. This CSO will remain open for Calls indefinitely with an annual update, until terminated. This CSO can result in the award of Contracts or Other Transaction Agreements (OTAs) as codified in 10 U.S.C. §2371b (section 2371b). The Government further reserves the right to award all, part, or none of the proposals received. All Calls issued under this CSO will include specific instructions including dates and times, Topics, evaluation criteria and proposal instructions to potential offerors. While each offeror shall adhere to this CSO, the Calls may contain additional information the offeror shall adhere to. Each Call will be announced on the Government Point of Entry SAM.gov, https://sam.gov/content/home/, other media may be used as well, and may result in the award of a FAR Part 12 contract or an OTA, if deemed in the Governments best interest. The Calls will contain broadly defined Topics and more specific problem topic interest areas. While these Topics are geared toward meeting requirements at JBSA, the Government reserves the right to award contracts under this CSO to meet Government requirements at other locations with similar Topics and problem topic interest areas. Prospective offerors shall focus their solutions on the Topics issued under the CSO Calls.

All details of the CSO Call process shall be governed by this CSO unless otherwise noted or expanded upon in each Call. The Government is not obligated to make any awards as a result of this CSO and/or Calls issued under this CSO, and all awards are subject to the availability of funds and successful negotiations. The Government is not responsible for any monies expended by the offeror prior to the issuance of any contract award to include, but not limited to bid and proposal cost. The Government reserves the right to amend the solicitation requirements of this CSO and each subsequent Call.

Calls issued under this CSO may utilize either a one-step or a two-step process as outlined below.

Note: Do not include Proprietary, classified, confidential, or sensitive information in responses. The government intends to use contractor support personnel to intake and process whitepapers and proposals.

One-Step Process: This process consists of offerors whose proposed solutions in response to a CSO Call clearly meet the Government's needs expressed in the Topic(s), or more specific interest areas via a favorable proposal evaluation. The offeror will receive an immediate notice that they have been selected for award or selected to provide more information that may subsequently lead to an award; or

Two-Step process:

1. Offerors must submit a white paper and or a proposal for evaluation in response to a posted CSO Call. Once the evaluations are complete, all offerors will be notified as to whether or not they have been selected to move to step-two of the CSO Evaluation Process. Submission of a White Paper that results in an unfavorable evaluation will not be considered further.

2. Proposals will only be accepted from Offerors who upon review of Phase I submissions, are invited by the Contracting Officer to submit a full proposal. In order to be considered for award under Phase II, offerors must be registered in the System for Award Management (SAM) at www.sam.gov, be considered responsible within the meaning of FAR Part 9.1, Responsible Prospective Contractors, have a satisfactory performance record, and otherwise be eligible for award based on federal law and regulation.

II. AREAS OF INTEREST

TOPIC: 001, ELECTROMAGNETIC WARFARE (EW)

The 16th AF seeks innovative and novel material solutions for EW. EW includes electromagnetic attack (EA), electromagnetic warfare support (ES), and electromagnetic protection (EP). All three contribute to operational success across the operational environment. Proper employment of EW capabilities produces the effects of detection, denial, deception, disruption, degradation, exploitation, destruction, and protection. Additionally sought is EW data fusion, advanced signal processing, cognitive techniques such as artificial intelligence (AI)/machine learning (ML), EA waveform generation, EW battle management (BM), and other advanced concepts. Capabilities to sense and provide a near instantaneous interpretation of the EM spectrum data and the information environment in real-time to characterize, identify change, and create predictions with minimal human interaction.

TOPIC: 002, INFORMATION OPERATIONS (IO)

The 16th AF seeks innovative and novel material solutions for IO. This topic includes: the ability to affect adversary and potential adversary decision making with the intent to ultimately affect their behavior in ways that help achieve friendly objectives. The decision making process exists within the information environment and consist of three targetable dimensions: 1) informational, 2) physical, and 3) cognitive. The information environment spans the content and delivery of the information used by the decision maker. Within the context herein, data becomes information once someone applies meaning to any data element.

TOPIC: 003, ELECTROMAGNETIC DIGITAL COMMUNICATIONS

Electromagnetic digital communications encompasses free space propagation of information and data elements through air and/or space, that may transition to terrestrial ground links. The offeror in ideal would have a method to achieve universal access no matter the networks' or material solution given basic information regarding the networks' physical configuration and protocol. Offerors may also propose methods for positively or negatively affecting (i.e. deny, disrupt, degrade, deceive) electromagnetic digital communications.

TOPIC: 004, MULTI-DOMAIN INNOVATION

The 16th AF seeks innovative and novel material solutions for innovation in integrated multi-domain operations. This topic includes multiple systems in one domain generating effects and/or supporting operations in another domain. For the purposes of this CSO, multi-domain operations capitalize on deep integration of EW, cyberspace, ISR, and Information Operations to generate effects within the tactical environment that result in operational and strategic results. Multi-domain operations often rely heavily on automated functions and managers due to the tight synchronization required for successful execution to present multiple dilemmas to overwhelm an adversary at multiple levels from machine to higher leadership that requires seamless, dynamic and continuous integration of capabilities generating effects in and from all domains.

TOPIC: 005, WEATHER AND CLIMATE RESOURCES

The 16th AF seeks innovative and novel solutions to include the following:

- Provide around-the-clock analyses, forecasts, warnings, and aircrew mission briefings 2)
- Deliver timely, relevant and specialized terrestrial, space and climatological global environmental intelligence
- Develop, evaluate, exploit and implement new tactics, techniques, procedures and technologies across Air Force Weather to enhance the effectiveness of Air Force

TOPIC: 006, INTELLIGENCE, SURVEILLANCE AND RECONNAISSANCE (ISR)

The 16th AF seeks innovative and novel solutions to include the following:

- Enable ISR Airmen to develop a comprehensive picture of the operational environment and, in coordination with the entire intelligence enterprise, develop a better understanding of potential adversaries' intentions and capabilities
- Improve collection, and integration of information derived from space, cyberspace, human, and open sources
- Advance capabilities in space, cyber, and human-derived intelligence, analytic expertise and tools

TOPIC: 007, AIR FORCE INFORMATION NETWORK (AFIN)

The 16th AF seeks innovative and novel solutions to include the following:

- Advance capabilities to install, support and prevent intrusion of systems
- Advance capabilities of the AF global communications network infrastructure
- Advance capabilities of AF computer hardware and software
- Secure, configure, operate, extend, maintain, and sustain DoD (AF) cyberspace and to create and preserve the confidentiality, availability, and integrity of the AFIN

TOPIC: 008, GEOPHYSICS and SEISMOLOGY

The 16th AF seeks innovative solutions for analytical techniques in seismology, hydroacoustics, and infrasonic phenomenology. This topic includes:

- Application of the use of lidar, digital imagery and cartographic products, such as geology maps, into event analysis.
- Application of the use of infrasound data for treaty monitoring purposes.
  - Modeling signal propagation through realistic atmospheres
  - Event location estimation with low uncertainty
  - Yield estimation with low uncertainty
  - Event discrimination
  - Evaluation of techniques for wind noise suppression with existing and experimental sensor technologies
- Modeling of seismic, infrasound and hydroacoustic signal-source scaling for a variety of sources or source conditions.

        - Fully contained explosions, underground or underwater
        - Partially contained explosions, underground or underwater
        - Surface explosions, over ground or water
        - Atmospheric explosions, over ground or water
        - Frequency modulated sources or Doppler signals from air or ground vehicles
    - Wave propagation and attenuation modeling with Full Waveform Earth Models
        - Improve the prediction of waveforms and amplitudes at local and regional distances
        - Use of cepstral analysis for event depth, multiple event identification
    - Managing ingestion and flow of multiple data sources for increased capabilities
        - Handle secure ingestion of data from multiple external sources
        - Handle movement of data across a guard to higher level networks
        - Including auditing of data flows
        - Data conversion between data formats
    - Application of new innovation sensors, improve qualification methods for new and existing sensors
    - Innovative hardware and software solutions for modern, flexible, and sustainable geophysical data acquisition systems and data processing; this can include (but is not limited to) new technologies to improve workstation processing, video display capabilities, and long-term data storage and retrieval.

TOPIC: 009, AIR AND SPACE

The 16th AF seeks innovative and novel solutions for the collection and analysis of signal observables from space, atmospheric and surface nuclear detonations (NUDET) that directly leads to operational technology. This topic includes:

- Modeling of any and all phenomenologies and processes between and including the NUDET or surrogate source and the recorded signal produced by the sensor detecting front end.
    - Modeling of the NUDET or surrogate signal generation, including interaction with its environment that results in a transmitted and detectable signal. Any and all phenomenologies are of interest (e.g., optical, RF, neutron, gamma-ray, x-ray).
    - Signal transport from the transmitted signal(s) to the signal arrival at the detecting sensor element. All intervening media are of interest (e.g., atmosphere, clouds, ionosphere, and space environment).
    - Conversion of the detectable signal at the sensor's front end into a data signal that can be processed (e.g., sensing element design, physics and engineering).
- Development and demonstration of computing environments and computational techniques for optimal NUDET signal analysis that are or can be used in an operational environment (e.g., High Performance Computing (HPC) applications, physics-based algorithms, and artificial intelligence (AI) tools to assist operational analysis).
- Translation of analysis algorithms and techniques into executable code.
- Implementation of sustainable, common ground architecture to rapidly integrate new detection platforms.

TOPIC: 010, DIRECTED ENEGRY WEAPON (DEW)

The 16th AF seeks innovative solutions for emerging DEW technologies in all domains. This topic includes:

- Advanced imaging and sensing, pointing and tracking systems, data collection and analysis, and modeling and simulation of optical and laser technologies.
- Develop methods and tools to enable and improve the analysis of multi-phenomena data sets.
- Enhanced tools and methods for clutter suppression, coherent processing and for extracting low

level signals of interest.

TOPIC: 011, NUCLEAR AND NUCLEAR-RELATED MATERIALS ANALYSIS

The 16th AF seeks innovative solutions to support trace-level to high-activity analyses of environmental samples for radionuclides. This topic includes:

- Operational laboratory and field methods to monitor potential radionuclide releases of "zero yield" events in order to confirm that nuclear fission did not occur during a foreign weapon test
- Detection of lower concentrations of radionuclides
- Nuclear debris collection

TOPIC: 012, MATERIALS MODELING AND ANALYSIS

The 16th AF seeks innovative solutions to support advanced meteorological and environmental modeling efforts. This topic includes:

- Advanced meteorological modeling tools and techniques, transport and dispersion modeling techniques, uncertainty estimation, and high performance computing
- Meteorological model development, advanced weather data assimilation techniques, ensemble modeling and uncertainty estimation techniques, transport and dispersion modeling, and high-performance computational techniques
- Advanced simulation and modeling techniques, environmental fate of compounds of interest, the identification of environmental constituents that may have an impact on US personnel or the performance of equipment and issues related to homeland defense
- Transport and dispersion modeling, computational chemistry, and process engineering

TOPIC: 013, ENTERPRISE WIDE INTEGRATION AND ARCHITECTURE MODERNIZATION

The 16th AF seeks innovative solutions to support the integration of data across disparate monitoring phenomenologies and modernization of hardware/software architectures. This topic includes:

- New solutions to integrate data access and discoverability across varying monitoring phenomenologies to lower detection thresholds and/or increase efficiency of current operations.
- Technologies to modernize hardware/software architectures or implement improved software design and accrediting processes to more flexibly meet mission needs.

TOPIC: 014, ENTERPRISE ASSET AND LIFECYCLE MAMAGEMENT IMPROVEMENTS

The 16th AF seeks innovative solutions that can provide enterprise wide asset management visibility as well as improve our lifecycle management capabilities. This topic includes:

- Increase accuracy of forecasting of requirements and scheduling of procurements through the use and exploitation of supply chain demand data
- Supply chain management, specifically: Automated systems to reduce/eliminate inefficiency, improve asset control, decrease touchpoints and minimize inventory
- Automated identification and reporting of components and systems with substandard reliability

TOPIC: 015, HIGH PERFORMANCE AND SECURE CLOUD COMPUTING AND DEV OPS SOLUTIONS FOR DATA INGESTION, EXPLOITATION, AND DISSEMINATION

The 16th AF seeks the following for this topic:

- Offer the ability to process data and perform complex calculations at high speeds, addressing the three critical HPC components (Computation, Network, and Storage).
- Utilize parallel processing to perform modeling, simulation and other scientific computing tasks.
- In a secure connection and networking environment, offer cloud computing that delivers the following hosted services over the Internet: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS).
- Software architecture and applications that can be utilized in the continuous development, continuous integration, and continuous delivery of software products that are being developed in an Agile software developing environment.
- Machine Learning (ML) and Artificial Intelligence (AI) Solutions.

TOPIC: 016, DEFENSE CYBERSPACE OPERATIONS (DCO) & DEFENSE CYBERSPACE OPERATIONS-INTERNAL DEFENSE MEASURES (DCO-IDM)

The 16th AF seeks the ability to utilize blue cyberspace capabilities and protect data, networks, cyberspace-enabled devices, and other designated systems by defeating on-going or imminent malicious cyberspace activity. This distinguishes DCO missions, which defeat specific threats that have bypassed, breached, or are threatening to breach security measures, from DODIN operations, which endeavor to secure DoD cyberspace from all threats in advance of any specific threat activity. Detect/defend actions that occur within the defended network or portion of cyberspace and dynamically reconfirm or reestablish the security of degraded, compromised, or otherwise threatened DoD cyberspace to ensure sufficient access to enable military missions.

III. CYBERSECURITY REQUIREMENTS AND SAM REGISTRATION

(https//www.SAM.gov/)

Cybersecurity Requirements

Small businesses looking to contract with the U.S. Department of Defense (DoD) will have to show the ability to safeguard their systems and data. Each DoD request for proposal will list a Cybersecurity Maturity Model Certification (CMMC) level required to bid the work.
To help small businesses with the tools and training to meet this standard, DoD developed Project Spectrum, a free platform that:

- Assists in CMMC certification
- Provides tools and training for cybersecurity awareness
- Educates users on risk management
- Helps small businesses install or boost cybersecurity hygiene

Visit Project Spectrum to sign up, learn about what you need to get certified, or to complete a self-assessment of your company's cyber readiness.

Unique Entity Identifier

Before you can bid on government proposals, you need to get a Unique Entity Identifier (UEI). A UEI is a unique 12-character, alpha-numeric value. You will receive a UEI when you register with SAM at SAM.gov. Businesses no longer have to go to a third-party website to obtain their identifier (DUNS number). This transition allows the government to streamline the entity identification and validation process, making it easier and less burdensome to do business with the federal government.

If your entity is already registered with SAM, your UEI has already been assigned to you. Learn how to view your UEI within SAM at the Federal Service Desk. Refer to the Guide to Getting a UEI if you want to get a UEI for your organization. For more information see, https://business.defense.gov/Work-with-us/Guide-to-working-with-DoD/

IV. PRICE DETERMINATION: FAIR AND REASONABLE

Price shall be considered to the extent appropriate, but at a minimum price will be determined fair and reasonable. The Government must determine the price fair and reasonable prior to award using the procedures at DFARS subpart 212.209. The contracting officer will use market research as the primary method to determine the price fair and reasonable. The Contracting Officer may request information from the offeror regarding recent purchase prices paid by the Government and/or commercial customers for the same or similar commercial items. Specific pricing requirements will be stated in CSO Calls.

V. DEFINITION OF TERMS

CSO with Calls: This allows for publication of an umbrella CSO solicitation that contains overarching information but does not request white papers or proposals. The CSO functions as a framework identifying the technical areas and giving the basic terms and administrative information of the CSO. Open and/or Closed Calls may be issued under this CSO to request submission of white papers and/or proposals. Calls may be issued at any time during the open period of the CSO. Each Call will be tailored to best fit the acquisition approach identified by the Government. The Calls may also include specific terms that apply to that Call such as further technical details and any pertinent clauses. Proposals and/or white papers are submitted to the Government only when Calls under this CSO are published.

Open Period Calls: This type of Call allows for white paper and/or proposal submissions at any time within a specified period as set forth in the Open Period Call.

Closed Calls: This type of Call allows for white paper and/or proposal submissions at a specified date and time as set forth in each Closed Call. This process and the dates associated with it are more structured than Open Period Calls.

Topics: These are specific requests for innovative solutions the Government will describe in each Call. An offeror's innovative solution(s) should focus on addressing specific Topics.

White Paper: A brief (usually 2-5 pages) summary of the proposed technical approach with an accompanying rough-order-of-magnitude (ROM) price.

Innovative: This refers to any technology, process, or method, including research and development that is new as of the date of proposal submission; or any new application of an existing technology, process or method as of the proposal date.

Nontraditional Defense Contractor: As defined in 10 U.S.C. § 2302(9) as an entity that is not currently performing and has not performed, for at least the one-year period preceding the solicitation of sources by the DoD for the procurement or transaction, any contract or subcontract for the DoD that is subject to full coverage under the cost accounting standards prescribed pursuant to 41 U.S.C. § 1502 and the regulations implementing such section. This includes all small business concerns under the criteria and size standards in 13 C.F.R. Part 121.

Small Business Concern: Defined in the Small Business Act (15 U.S.C. 632) and 13 CFR Part 121.